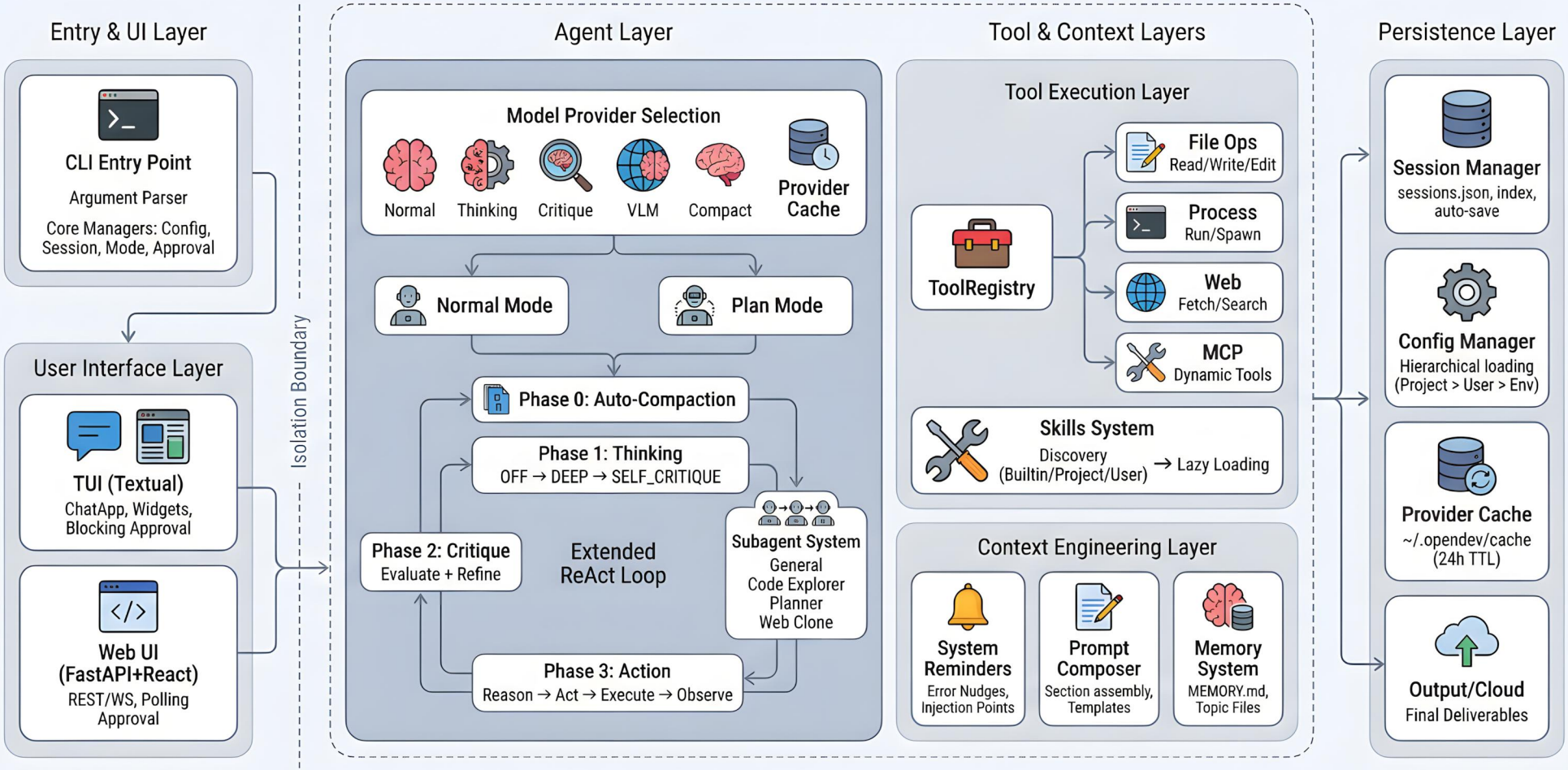
Entry & UI Layer
CLI Entry Point
Argument Parser
Core Managers: Config, Session, Mode, Approval
User Interface Layer
TUI (Textual)
ChatApp, Widgets, Blocking Approval
Web UI (FastAPI+React)
REST/WS, Polling Approval
Isolation Boundary
Agent Layer
Model Provider Selection
Normal
Thinking
Critique
VLM
Compact
Provider Cache
Normal Mode
Plan Mode
Phase 0: Auto-Compaction
Phase 1: Thinking
OFF → DEEP → SELF_CRITIQUE
Phase 2: Critique
Evaluate + Refine
Extended ReAct Loop
Subagent System
General
Code Explorer
Planner
Web Clone
Phase 3: Action
Reason → Act → Execute → Observe
Tool & Context Layers
Tool Execution Layer
ToolRegistry
File Ops
Read/Write/Edit
Process
Run/Spawn
Web
Fetch/Search
MCP
Dynamic Tools
Skills System
Discovery (Builtin/Project/User)
→ Lazy Loading
Context Engineering Layer
System Reminders
Error Nudges, Injection Points
Prompt Composer
Section assembly, Templates
Memory System
MEMORY.md, Topic Files
Persistence Layer
Session Manager
sessions.json, index, auto-save
Config Manager
Hierarchical loading (Project > User > Env)
Provider Cache
~/.opendev/cache (24h TTL)
Output/Cloud
Final Deliverables